AIM:

To write a program to find the solution of a matrix using Gaussian Elimination.

Equipments Required:

1. Hardware – PCs
2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

Algorithm

```
1.Start the program
2.Import the necessary libraries(numpy,scipy.linalg)
3.Define the matrix using numpy
4.Use lu(),lu_solve(),lu_factor() to get the solutions
5.End the program
```

Program:

```
Program to find the solution of a matrix using Gaussian Elimination.
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import os
os.environ["OPENBLAS_NUM_THREADS"]="1"

import numpy as np
import sys
n = int(input())
a = np.zeros((n,n+1))
x = np.zeros(n)
for i in range(n):
    for j in range(n+1):
        a[i][j] = float(input())
for i in range(n):
    if a[i][i] == 0.0:
        sys.exit('Divide by zero detected!')
    for j in range(i+1, n):
        ratio = a[j][i]/a[i][i]
        for k in range(n+1):
            a[j][k] = a[j][k] - ratio * a[i][k]
```
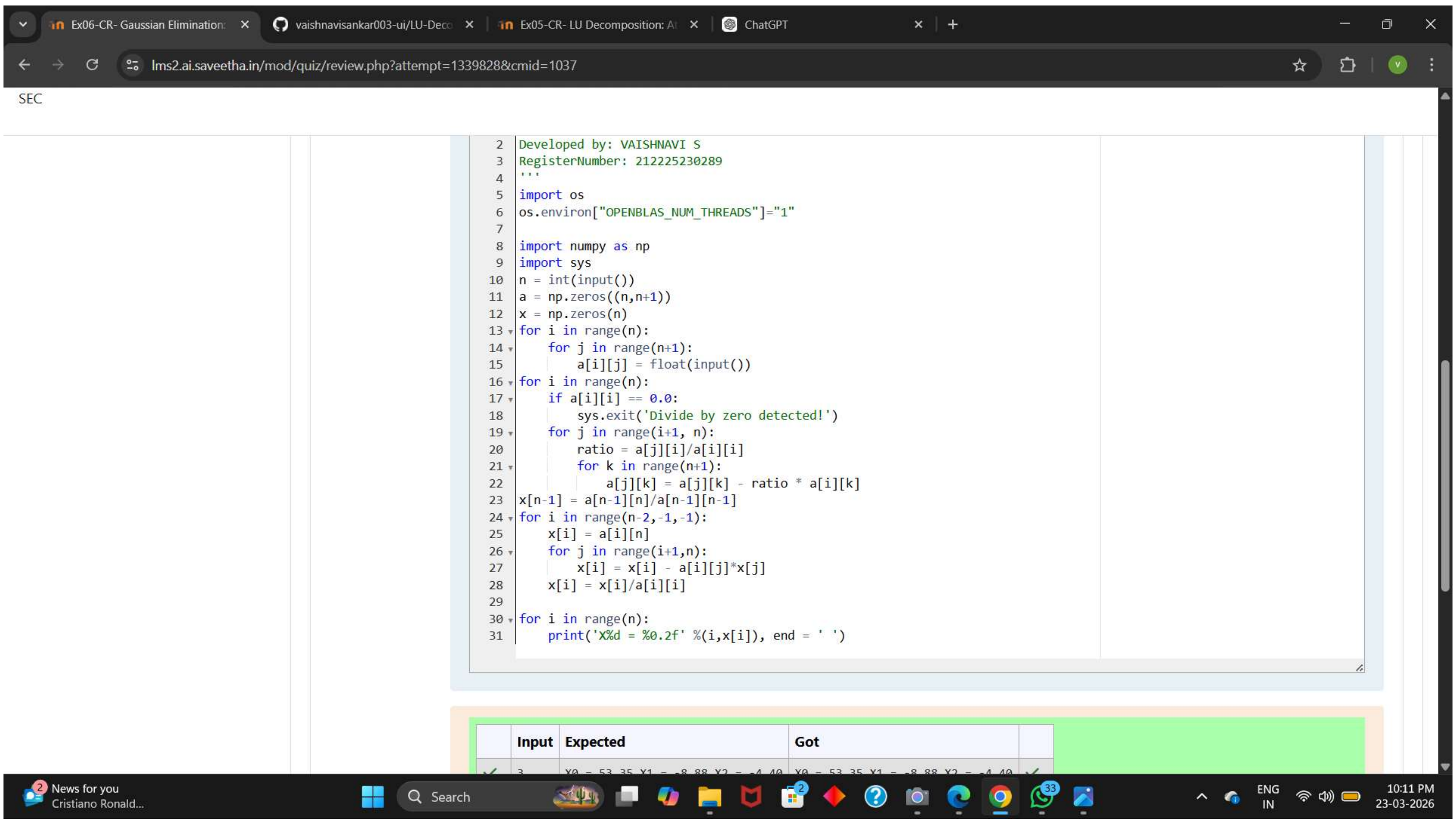

```
x[n-1] = a[n-1][n]/a[n-1][n-1]
for i in range(n-2,-1,-1):
    x[i] = a[i][n]
    for j in range(i+1,n):
        x[i] = x[i] - a[i][j]*x[j]
    x[i] = x[i]/a[i][i]

for i in range(n):
    print('X%d = %0.2f' %(i,x[i]), end = ' ')
```

Output:



Result:

Thus the program to find the solution of a matrix using Gaussian Elimination is written and verified using python programming.